

Package ‘manMetaVAR’

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Title Multivariate Meta-Analysis of Vector Autoregressive Model
Coefficients

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BugReports <https://github.com/jeksterslab/manMetaVAR/issues>

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coef.manmetavar.dtvvar	<i>Parameter Estimates (FitDTVAR)</i>
------------------------	---------------------------------------

Description

Parameter Estimates (FitDTVAR)

Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
```

```

    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

```

Arguments

object	Object of class <code>manmetavar.dtvvar</code> .
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
vanishing_theta	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
theta_tol	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
...	additional arguments.

Value

Returns a list of vectors of parameter estimates.

Author(s)

Ivan Jacob Agaloos Pesigan

```
coef.manmetavar.metavar
```

Parameter Estimates (FitMetaVAR)

Description

Parameter Estimates (FitMetaVAR)

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)
```

Arguments

object	Object of class manmetavar.metavar.
...	additional arguments.

Value

Returns a vector of estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

```
Compress
```

Compress Replication

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

Dynamics

Process Dynamics

Description

Process Dynamics

Usage

Dynamics(dynamics)

Arguments

dynamics 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

See Also

Other Data Generation Functions: [GenData\(\)](#)

Examples

```
## Not run:
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)

## End(Not run)
```

FitDTVAR

Fit the Model using the fitDTVARMxID Package

Description

The function fits the model using the [fitDTVARMxID::fitDTVARMxID](#) package.

Usage

FitDTVAR(data, ncores = NULL, seed = NULL)

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitMLVAR\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitMetaVAR

Multivariate Meta-Analysis using the metaVAR Package

Description

The function performs multivariate meta-analysis using the [metaVAR::metaVAR](#) package.

Usage

```
FitMetaVAR(fit, ncores = NULL)
```

Arguments

fit	R object. Output of the FitDTVAR() function.
ncores	Positive integer. Number of cores to use.

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```

FitMLVAR

Fit the Model using the mlVAR Package

Description

The function fits the model using the [mlVAR::mlVAR](#) package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMLVAR(data = data)
print(fit)
summary(fit)
```

```
## End(Not run)
```

GenData	<i>Simulate Data</i>
---------	----------------------

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid)
```

Arguments

taskid Positive integer. Task ID.

See Also

Other Data Generation Functions: `Dynamics()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)

## End(Not run)
```

params	<i>Simulation Parameters</i>
--------	------------------------------

Description

Simulation Parameters

Usage

```
params
```


Format

A dataframe with 25 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occasions.

dynamics Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

Author(s)

Ivan Jacob Agaloos Pesigan

`plot.manmetavar.data` *Plot Method for an Object of Class manmetavar.data*

Description

Plot Method for an Object of Class `manmetavar.data`

Usage

```
## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

<code>x</code>	Object of class <code>manmetavar.data</code> .
<code>...</code>	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

print.manmetavar.data *Print Method for an Object of Class manmetavar.data*

Description

Print Method for an Object of Class manmetavar.data

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)
```

Arguments

x	Object of class manmetavar.data.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

print.manmetavar.dtvvar
Print Method (FitDTVAR)

Description

Print Method (FitDTVAR)

Usage

```
## S3 method for class 'manmetavar.dtvvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
```

```

    digits = 4,
    ...
)

```

Arguments

x	Object of class <code>manmetavar.dtvvar</code> .
means	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
vanishing_theta	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
theta_tol	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

```
print.manmetavar.metavar
```

Print Method (FitMetaVAR)

Description

Print Method (FitMetaVAR)

Usage

```
## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)
```

Arguments

x	Object of class manmetavar.metavar.
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

```
print.manmetavar.mlvar
```

Print Method (FitMLVAR)

Description

Print Method (FitMLVAR)

Usage

```
## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

Arguments

x	Object of class manmetavar.mlvar.
show	Tables to show.
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim

Simulation Replication

Description

Simulation Replication

Usage

Sim(taskid, repid, output_folder, overwrite, integrity, seed)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR

Simulation Replication - FitDTVAR

Description

Simulation Replication - FitDTVAR

Usage

SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR

Simulation Replication - FitMetaVAR

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMLVAR	<i>Simulation Replication - FitMLVAR</i>
-------------	--

Description

Simulation Replication - FitMLVAR

Usage

SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
-------	-----------------------------

Description

Simulation File Name

Usage

SimFN(output_type, output_folder, suffix)

Arguments

- output_type Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var".
- output_folder Character string. Output folder.
- suffix Character string. Output of manCTMed:::SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
------------	---

Description

Simulation Replication - GenData

Usage

SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

- taskid Positive integer. Task ID.
- repid Positive integer. Replication ID.
- output_folder Character string. Output folder.
- seed Integer. Random seed.
- suffix Character string. Output of manCTMed:::SimSuffix().
- overwrite Logical. Overwrite existing output in output_folder.
- integrity Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj	<i>Simulation Project Name</i>
---------	--------------------------------

Description

Simulation Project Name

Usage

SimProj()

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar.data	<i>Summary Method for an Object of Class manmetavar.data</i>
-------------------------	--

Description

Summary Method for an Object of Class manmetavar.data

Usage

```
## S3 method for class 'manmetavar.data'  
summary(object, ...)
```

Arguments

object	Object of class manmetavar.data.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar.dtvvar
<i>Summary Method (FitDTVAR)</i>

Description

Summary Method (FitDTVAR)

Usage

```
## S3 method for class 'manmetavar.dtvvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)
```

Arguments

object	Object of class manmetavar.dtvvar.
means	Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.

theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar.metavar

Summary Method (FitMetaVAR)

Description

Summary Method (FitMetaVAR)

Usage

```
## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)
```

Arguments

object	Object of class manmetavar.metavar.
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

```
summary.manmetavar.mlvar
```

Summary Method (FitMLVAR)

Description

Summary Method (FitMLVAR)

Usage

```
## S3 method for class 'manmetavar.mlvar'
summary(
  object,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

Arguments

object	Object of class manmetavar.mlvar.
show	Tables to show.
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

```
vcov.manmetavar.dtvvar
```

Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)

Description

Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)

Usage

```
## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)
```

Arguments

<code>object</code>	Object of class <code>manmetavar.dtvvar</code> .
<code>alpha</code>	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
<code>beta</code>	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
<code>nu</code>	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
<code>psi</code>	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
<code>theta</code>	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
<code>converged</code>	Logical. Only include converged cases.
<code>grad_tol</code>	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
<code>hess_tol</code>	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
<code>vanishing_theta</code>	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
<code>theta_tol</code>	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
<code>...</code>	additional arguments.

Value

Returns a list of sampling variance-covariance matrices.

Author(s)

Ivan Jacob Agaloos Pesigan

vcov.manmetavar.metavar

Sampling Covariance Matrix of the Parameter Estimates (Fit-MetaVAR)

Description

Sampling Covariance Matrix of the Parameter Estimates (FitMetaVAR)

Usage

```
## S3 method for class 'manmetavar.metavar'  
vcov(object, ...)
```

Arguments

object	Object of class manmetavar.metavar.
...	additional arguments.

Value

Returns the sampling variance-covariance matrix of the estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

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